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Academic Positions

- 2026 Aug - present **Northwestern University**, Department of Physics and Astronomy, Evanston, USA
- **Research Associate**, Advisor: Prof. Ian Low
- 2025 Jun - 2026 Jul **CERN**, Department of Theoretical Physics, Geneva, Switzerland
- **Scientific Associate**
- 2022 Jun - 2025 May **Massachusetts Institute of Technology (MIT)**, Center of Theoretical Physics & Institute of Artificial Intelligence and Fundamental Interactions, Cambridge, USA
- **Postdoctoral Fellow**, Advisor: Prof. Dr. Jesse Thaler
- 2020 Oct - 2022 May **Scuola Internazionale Superiore di Studi Avanzati (SISSA)**, Department of Physics, Trieste, Italy
- **Postdoctoral Fellow**, Advisor: Dr. David Marzocca

Education

- 2016 Dec - 2020 Sep **Universität Zürich**, Zürich, Switzerland
- **Ph.D. in Physics**, Advisor: Prof. Dr. Gino Isidori
- 2015 Feb - 2016 Nov **Eidgenössische Technische Universität Zürich (ETH)**, Zürich, Switzerland
- **M.Sc. in Physics**, Advisor: Prof. Dr. Gino Isidori
- 2011 Oct - 2014 Sep **Technische Universität Dortmund**, Dortmund, Germany
- **B.Sc. in Physics**, Advisor: Prof. Dr. Emmanuel A. Paschos

Awards and Grants

- 2023 **SNSF Return Grant** (P5R5PT_222350), Switzerland
- 2022 **SNSF Postdoc.Mobility Grant** (P500PT_203156), Switzerland
- 2020 **INFN Assegno di Ricerca Fellowship** (2017L5W2PT), Italy
- 2018 **Invisibles Plus Exchange Scholarship**, Lawrence Berkeley Lab, USA
- 2016 **ETH & TokyoTech Exchange Scholarship**, Tokyo, Japan
- 2016 Global Essay Competition Finalist, 46th St. Gallen Symposium, Switzerland

Continuing Education

- 2024 Feb Brookhaven National Laboratory, EIC Visitor Program, Upton, USA
- 2023 Jun Jefferson Lab, Quantum Computing Bootcamp, Newport News, USA
- 2022 Aug IAIFI Summer School & Workshop, Cambridge, USA

2020 Aug	EPFL, Machine Learning in High Energy Physics, Geneva, Switzerland (online)
2018 May	Higgs Centre for Theoretical Physics, Edinburgh, United Kingdom
2017 Jul	Institute for Advanced Study, Prospects in High Energy Physics, Princeton, USA

Teaching Experience

Workshop tutorial leader, IAIFI Summer School

2024 Aug	Representation/Manifold Learning (Prof. M. Weber)
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Teaching assistant, Universität Zürich

2019 Fall	Quantum Field Theory III (Prof. G. Isidori)
2019 Spring	Advanced Field Theory (Dr. Lazopoulos)
2018 Fall	Quantum Field Theory I (Prof. A. Gehrmann-De Ridder),
2018 Spring	Quantum Mechanics II (Prof. A. Signer)
2017 Fall	General Relativity (Prof. G. M. Graf)
2017 Spring	Quantum Field Theory II (Prof. N. Beisert)

Replacement lecturer, Universität Zürich

2019 Fall	Quantum Field Theory III (Prof. G. Isidori)
2018 Spring	Quantum Mechanics II (Prof. A. Signer)

2015	Seminar for Didactics , ETH Department of Education and Technology
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2015-2016	Teaching assistant , ETH, undergraduate courses
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Mentoring

Graduate student research advisor

2026-present	Alexander G. Stapleton (Queen Mary)
2025-present	Alexandra P. Klipfel (MIT)
2025-present	Thomas Driscoll (U. of Oregon)
2024-present	Kate Richardson (MIT)
2024-2025	Chaja Baruch (Technion)
2023-2025	Ta'el Coren (Technion)
2023-2025	Sean Benevedes (MIT)
2023	Pamela Pajarillo (MIT)
2022	Miguel Vanvlasselaer (SISSA, now: postdoc at U. Barcelona)

Undergraduate student research advisor

2026-present	Radu Murariu (ETH Zurich)
2025-present	Dylan Perez (Caltech)
2025-present	Angelos Sirbu (University of Crete)
2023-present	Sabina Tomasicchio (U. of Seville, now Ph.D. at UMass Amherst)
2023-2024	Victor Samuel Pérez Díaz (Universidad del Rosario, now: Ph.D. at NYU)
2023 Jun-Oct	Alicia Mand (University of Wisconsin-Madison, Ph.D. at University of Wisconsin-Madison)

- 2023-2024 ○ Dhruv Kumar (IIT Guwahati, now: intern at CADSL IIT Bombay)
 - 2023 Jun-Oct ○ Acchhyut Jolly (BITS Pilani)
 - 2023 Jun-Oct ○ Soham Sanyashiv (IISER Kolkata, now: intern at TIFR, Mumbai)
 - 2023 Jun-Oct ○ Gokhula Prasad (The American College, Madurai, now: Ph.D. at NIT Tiruchirappalli)
 - 2023 Jun-Oct ○ Abhay Singh Rawat (HNBGU, now: Ph.D. at HNBGU)
- International Baccalaureate Diploma tutor**
- 2016 ○ Dustin Fichmann (United World College Costa Rica, now: Scientist at World Data Lab)

Conference and Workshop Talks

“Fundamental Interactions from Quantum Information”

- 2026 Jul **Invited plenary:** SynQretism 2026, Chania, Greece
- 2026 May **Invited:** Annual Meeting of the LHC Top Physics Working Group, CERN, Geneva, Switzerland
- 2025 Dec **Invited workshop:** Quantum Information and Computation for Particle Physics Day, Complutense University of Madrid, Madrid, Spain
- 2025 Sep **Invited:** CERN TH Institute on Quantum Simulation & Computation in High-Energy Physics, Geneva, Switzerland
- 2025 Apr **Invited plenary:** Quantum Observables for Collider Physics, Florence, Italy
- 2025 Mar **Invited plenary:** Moriond Electroweak Interactions & Unified Theories, La Thuile, Italy
- 2025 Feb **Invited plenary:** KEK Theory Meeting, Tsukuba, Japan

“From Neurons to Neutrons: An interpretable AI for nuclear physics”

- 2025 Dec **Invited plenary:** Christmas Symposium on Theoretical Physics, Thessaloniki, Greece
- 2025 Oct Contributed: 16th European Research Conference on Electromagnetic Interactions with Nucleons and Nuclei, Paphos, Cyprus
- 2025 Aug **Invited plenary:** IAIFI Summer Workshop, Cambridge, USA
- 2024 Aug Contributed: XVIth Quark Confinement and the Hadron Spectrum, Cairns, Australia
- 2024 Jun **Peer-reviewed poster:** ICML 2024, Vienna, Austria
- 2023 Aug Poster: AI for Science, Chicago

“Charting the Universe with Primordial Black Holes”

- 2026 Jul **Invited plenary:** Syncretism 2026, Chania, Greece
- 2026 Jun **Invited plenary:** New Horizons in Primordial Black Hole Physics, CERN, Switzerland
- 2025 Aug Contributed: Black Holes & Cosmology, Reykjavík, Iceland

	“New Physics at Colliders: The present, the future and the far future”
2024 Jul	Invited plenary: Light Dark World, Daejeon, Korea
	“BSM at the Muon Synchrotron Ion Collider: Let the MuSIC play!”
2025 Mar	Contributed: Empowering the New Vision in High Energy Physics, Aspen, USA
2024 Nov	Invited plenary: Uncovering New Laws of Nature at EIC, Upton, USA
	“Pulsar Timing Arrays gravitational waves and JWST early galaxies”
2024 Jun	Invited plenary: SynCRETism 2024, Rethymno, Greece
2024 Mar	Invited: Workshop on PBHs, MIT LIGO Lab, Cambridge, USA
2024 Mar	Contributed: Black Holes & Cosmology, Nassau, Bahamas
2023 Dec	Invited: Xmas Theoretical Physics Workshop, Athens, Greece
2023 Aug	Invited: Invisibles Workshop, Göttingen, Germany
	“Standard Model PDFs for the Muon Collider”
2023 Mar	Invited: LISHEP2023, Rio de Janeiro, Brazil
2022 Aug	Contributed: Vietnam Flavour Physics Conference, Quy Nhon, Vietnam
	“A Minimal Explanation of Flavour Anomalies”
2021 Sep	Contributed: Workshop on the Standard Model and Beyond, Corfu, Greece
	“Collider Signatures of Coannihilating Dark Matter”
2021 Jun	Contributed: Workshop on Axions, WIMPs, and WISPs, Trieste, Italy (online)
2020 Mar	Invited: La Thuile, Aosta Valley, Italy; canceled
	“Exploring the flavour structure of the high-scale MSSM”
2021 Aug	Contributed: SUSY International Conference on Supersymmetry and the Unification of Fundamental Interactions, Beijing, China
2017 Dec	Contributed: SUSY International Conference on Supersymmetry and the Unification of Fundamental Interactions, Mumbai, India
2017 Feb	Contributed: PRISMA Symposium "A Matter of Flavor", Mainz, Germany
	“The bridge between R-parity violating SUSY and flavoured Dark Matter”
2019 Jun	Contributed: PLANCK International Conference from the Planck Scale to the Electroweak Scale, Granada, Spain
2019 May	Contributed: SUSY International Conference on Supersymmetry and the Unification of Fundamental Interactions, Corpus Christi, USA
	Invited Seminar Talks & Colloquia
	“The Phenomenologist’s Guide to the Dark Universe”
2026 Mar	Deutsches Elektronen-Synchrotron DESY, Hamburg, Germany (colloquium)
2026 Feb	University of Pittsburgh, Pittsburgh, USA
2026 Feb	National Centre of Scientific Research “Demokritos”, Athens, Greece
2025 Jun	University of Crete, Heraklion, Greece (colloquium)

“Fundamental Interactions from Quantum Information”

2025 Oct University of Cyprus, Nicosia, Cyprus
2025 Oct Aristotle University, Thessaloniki, Greece
2025 Sep University of Oxford, Oxford, UK
2025 May Northwestern University, Evanston, Illinois
2025 Apr Indian Institute of Science, Bangalore, India (online)
2025 Jan Kavli IPMU, Kashiwa, Japan
2025 Jan Lawrence Berkeley National Laboratory, Berkeley, USA
2024 Dec UC San Diego, San Diego, USA

“From Neurons to Neutrons: An interpretable AI for nuclear physics”

2026 Jun University of Münster, Münster, Germany
2025 Oct CERN TH (Heavy-Ion Seminar), Geneva, Switzerland
2025 Oct Queen Mary University of London, London, UK

“Charting the Universe with Primordial Black Holes”

2026 Apr NICPB, Tallinn, Estonia
2025 May Kavli Institute for Cosmological Physics, Chicago, Illinois
2025 Feb Chinese Academy of Sciences, Beijing, China

“New Physics at Colliders: The present, the future and the far future”

2025 Jan University of Tokyo, Tokyo, Japan
2024 Sep University of Melbourne, Melbourne, Australia
2024 Mar Argonne National Laboratory, Lemont, USA
2024 Mar Brookhaven National Laboratory, Upton, USA

“Pulsar Timing Arrays gravitational waves and JWST early galaxies”

2023 Dec UC Berkeley, Berkeley, USA
2023 Dec Vrije Universiteit Brussel, Brussels, Belgium
2023 Oct Chinese Academy of Sciences, Beijing, China (online)

“Attracting the electroweak scale to a tachyonic trap”

2023 Mar Majorana-Raychaudhuri Seminars Series (online)
2023 Jan Technion, Haifa, Israel

“Standard Model PDFs for the Muon Collider”

2023 Mar Technion, Haifa, Israel
2023 Mar University of Chicago, Chicago, USA
2022 Sep MIT, Cambridge, USA

“The bridge between R-parity violating SUSY and flavoured Dark Matter”

2019 Sep	Fermilab, Batavia, USA
2019 Sep	Cornell, Ithaca, USA
2019 Aug	Lawrence Berkeley National Laboratory, Berkeley, USA
2019 Aug	UC Santa Cruz, Santa Cruz, USA

Academic Service and Scientific Societies

2026-2027	Lead Conference Organizer: Quantum Observables for Collider Physics 2027, Northwestern University
2025-2026	Lead Conference Organizer: Quantum Observables for Collider Physics 2026, CERN
2025-2026	Conference Organizer: PLANCK2026 & 6th EuCAPT Symposium, CERN
2023-2024	School & Workshop Organizer: IAIFI Summer School & Workshop 2024
2019-present	Journal referee: PRL, PRA, PRD, EPJC, Phys. Lett. B, JHEP, IJMPA
2023-present	Machine Learning Conference referee: NeurIPS, ICML, ML4PS
2025-present	Member: Greek Astrophysical Society
2025-present	Member: European Astrophysical Society
2025-present	Member: Asymmetry network
2018 Jan	Workshop organizer: Zurich Phenomenology Workshop
2018-2019	Organizer: Theoretical Physics Journal Club of Zürich

Public Outreach

2025-2026	CERN guide: Volunteer for guided tours of CERN facilities and local outreach events organized by Science Gateway; completed the Data Center Visitor Point course.
2022-2025	Committee member: Organizer and representative for IAIFI public outreach events.
2023 Oct	Public lecture: “Spot the Difference: AI vs Reality in Physics”, Cambridge Science Festival, Cambridge, USA
2023 Aug	Public lecture: “The Interplay between Physics and Artificial Intelligence”, Museum of Science, Boston, USA
2023 Jul	Public lecture: “Machine Learning Applications to Physics”, Remote Experience for Young Researchers, Berkeley, USA
2023 Jun	Invited industry talk: “AI technologies: the new frontier”, Industry Seminar FONTIS Beratung, Zürich, Switzerland

Languages

English (fluent), German (fluent), Greek (native).

Coding

Languages	C++, Python, Julia, Mathematica.
Packages	MadGraph5, Pythia8, Delphes3, FastJet3, micrOMEGAs, MontePython, COLOSSUS, CLASS,

PTArcade, CosmoLattice, OriginPro, Python libraries (PyTorch, NumPy, SciPy, scikit-learn, pandas, PySR, JAX, etc.), wandb.ai, Slurm.

Agentic AI LLM-based agents (OpenAI GPT, Anthropic Claude, and Ollama models: DeepSeek-R1, GPT-OSS, Qwen3), Ollama Python and JavaScript libraries, structured critic-actor workflows, tool-use/RAG, and evaluation pipelines.

Publications in Peer-Reviewed Journals

- ❖ See my [INSPIRE page](#) for a complete publication list, including work with 60+ distinct co-authors outside large collaborations.

38. “Explaining the Origin of the Universe’s Heavy Elements with AI: Precision Nuclear Input for r-process Nucleosynthesis”. S. G. Tomasicchio, **S. Trifinopoulos**, in preparation
37. “Learning Nuclear Structure with AI: Radii and Collectivity”. M. Williams, G. Giacalone, **S. Trifinopoulos**, in preparation
36. “Infrared Structure of Soft-Photon Decoherence in Fermion Scattering”. I. Low, Z. Yin, **S. Trifinopoulos**, in preparation
35. “The Moon as a Background-Free Probe of Exotic μHz Chirping Sources”. G. Franciolini, Y. Gouttenoire, A. J. Iovino, M. Vanvlasselaer, **S. Trifinopoulos**, in preparation
34. “Next-to-Leading-Order Electroweak Corrections to Quantum Observables in Lepton-Lepton Collisions”. S. Chang, T. Driscoll, **S. Trifinopoulos**, submitted for peer review • arXiv: 2608.27559 [hep-ph, quant-ph]
33. “Quantum Resources and Wigner Symmetry in Nucleon-Nucleon Scattering from Effective Field Theory”. I. Low, T. R. Richardson, **S. Trifinopoulos**, submitted for peer review • arXiv: 2606.17148 [nucl-th, quant-ph]
32. “When Does Critique Improve AI-Assisted Theoretical Physics? SCALAR: Structured Critic–Actor Loop for Agentic Reasoning”. V. Niarchos, C. Papageorgakis, A. G. Stapleton, **S. Trifinopoulos**, accepted after peer review at ICML 2026 “AI4Physics Workshop” • arXiv: 2605.06772 [cs.AI, hep-th]
31. “Neural Operators as Efficient Function Interpolators”. V. Niarchos, A. Sirbu, **S. Trifinopoulos**, submitted for peer review • arXiv: 2605.07792 [cs.LG, nucl-th]
30. “On Exclusive Coherent Production of Bosons in Electron-Proton Collisions”. R. Balkin, T. Coren, A. Jentsch, H. Liu, M. Ovchinnikov, Y. Soreq, **S. Trifinopoulos**, accepted in Phys. Rev. D • arXiv: 2604.08667 [hep-ph]
29. “Baryogenesis from Exploding Primordial Black Holes”. A. P. Klipfel, M. Vanvlasselaer, **S. Trifinopoulos**, D. I. Kaiser, submitted for peer review • arXiv: 2603.29024 [hep-ph, astro-ph.CO]
28. “Shocks from Exploding Primordial Black Holes in the Early Universe”. M. Vanvlasselaer, **S. Trifinopoulos**, A. P. Klipfel, D. I. Kaiser, submitted for peer review • arXiv: 2603.15746 [astro-ph.CO, astro-ph.HE]
27. “The Dark Side of the Moon: Listening for Scalar-Induced Gravitational Waves”. D. Blas, J. W. Foster, Y. Gouttenoire, A. J. Iovino, I. Musco, **S. Trifinopoulos**, M. Vanvlasselaer, Phys. Lett. B 879 (2026) 140635 • arXiv: 2602.12252 [astro-ph.CO]

26. “Braking protons at the EIC: from invisible meson decay to new physics searches”. R. Balkin, T. Coren, A. Jentsch, H. Liu, M. Ovchinnikov, Y. Soreq, **S. Trifinopoulos**, accepted in Phys. Rev. Lett. • arXiv: 2601.00068 [hep-ph]
25. “Large-Scale Structure Probes of the Post-Inflationary Axiverse”. M. Gorghetto, **S. Trifinopoulos**, G. Valogiannis, JCAP 06 (2026) 031 • arXiv: 2511.04734 [astro-ph.CO, hep-ph]
24. “Quantum Information at the Electron-Ion Collider”. K. Cheng, T. Han, **S. Trifinopoulos**, accepted in Phys. Rev. D • arXiv: 2510.23773 [hep-ph, quant-ph]
23. “Implications for Pulsar Timing Arrays of Sub-solar Black Hole Detections: From LVK to Einstein Telescope and Cosmic Explorer”. Y. Gouttenoire, **S. Trifinopoulos**, M. Vanvlasselaer, JCAP 02 (2026) 072 • arXiv: 2508.19328 [astro-ph.CO]
22. “Spin versus Magic: Lessons from Gluon and Graviton Scattering”. J. Gargalionis, N. Moynihan, **S. Trifinopoulos**, E. N. V. Wallace, C. D. White, and M. J. White, Phys. Rev. D 113 (2026) 1, 016007 • arXiv: 2508.14967 [hep-th, quant-ph]
21. “The DNA of nuclear models: How AI predicts nuclear masses”. K. A. Richardson, **S. Trifinopoulos**, M. Williams, accepted in Nature AI & Computing • arXiv: 2508.08370 [nucl-th, cs.ai]
20. “Effective Field Theory Constraints on Primordial Black Holes from the High-Redshift Lyman- α Forest”. M. M. Ivanov, **S. Trifinopoulos**, Phys. Rev. Lett. 136 (2026) 17, 171402 • arXiv: 2508.04767 [astro-ph.CO]
19. “Searching for exotic scalars at fusion reactors”. C. Baruch, P. J. Fitzpatrick, T. Menzo, Y. Soreq, **S. Trifinopoulos**, J. Zupan, JHEP 10 (2025) 215 • arXiv: 2502.12314 [hep-ph]
18. “New Physics at the Muon (Synchrotron) Ion Collider: MuSIC for several scales”. H. Davoudiasl, H. Liu, R. Marcarelli, Y. Soreq, **S. Trifinopoulos**, JHEP 03 (2025) 046 • arXiv: 2412.13289 [hep-ph]
17. “Flavor Patterns of Fundamental Particles from Quantum Entanglement?”. J. Thaler, **S. Trifinopoulos**, Phys. Rev. D 111 (2025) 5, 056021 • arXiv: 2410.23343 [hep-ph, quant-ph]
16. “Spontaneous symmetry breaking, gauge hierarchy and electroweak vacuum metastability”. S. Benevedes, T. Steingasser, **S. Trifinopoulos**, Phys. Rev. D 110 (2024) 7, 075019 • arXiv: 2408.10297 [hep-ph, astro-ph.CO]
15. “From Neurons to Neutrons: A Case Study in Interpretability”. O. Kitouni, N. Nolte, V. S. Pérez-Díaz, **S. Trifinopoulos**, M. Williams, ICML 2024 • arXiv: 2405.17425 [cs.lg, nucl-th]
14. “Scrutinizing the Primordial Black Holes Interpretation of PTA Gravitational Waves and JWST Early Galaxies”. Y. Gouttenoire, **S. Trifinopoulos**, G. Valogiannis, M. Vanvlasselaer, Phys. Rev. D 109 (2024) 12, 123002 • arXiv: 2307.01457 [astro-ph.CO]
13. “LePDF: Standard Model PDFs for High-Energy Lepton Colliders”. F. Garosi, D. Marzocca, **S. Trifinopoulos**, JHEP 09 (2023) 107 • arXiv: 2303.16964 [hep-ph]

12. “Cabibbo angle anomalies and oblique corrections: The remarkable role of the vectorlike quark doublet”. B. Belfatto, **S. Trifinopoulos**, Phys. Rev. D 108 (2023) 3, 035022 • arXiv: 2302.14097 [hep-ph]
11. “Attracting the Electroweak Scale to a Tachyonic Trap”. **S. Trifinopoulos**, M. Vanvlasselaer, Phys. Rev. D 107 (2023) 7, L071701 • arXiv: 2210.13484 [hep-ph]
10. “New physics in $b \rightarrow s\mu\mu$: FCC-hh or a muon collider?” A. Azatov, F. Garosi, A. Greljo, D. Marzocca, J. Salko, **S. Trifinopoulos**, JHEP 10 (2022) 149 • arXiv: 2205.13552 [hep-ph, hep-ex]
9. “Radiative effects in the scalar sector of vector leptoquark models”. R. Houtz, J. Pagès, **S. Trifinopoulos**, JHEP 08 (2022) 208 • arXiv: 2204.06440 [hep-ph]
8. “Displaced searches for light vector bosons at Belle II”. T. Bandyopadhyay, S. Chakraborty, **S. Trifinopoulos**, JHEP 05 (2022) 141 • arXiv: 2203.03280 [hep-ph]
7. “Collider signatures of coannihilating dark matter in light of the B-physics anomalies”. M.J. Baker, D. A. Faroughy, **S. Trifinopoulos**, JHEP 11 (2021) 084 • arXiv: 2109.08689 [hep-ph]
6. “From B-meson anomalies to Kaon physics with scalar leptoquarks”. D. Marzocca, **S. Trifinopoulos**, E. Venturini, Eur. Phys. J. C 82 (2022) 320 • arXiv: 2106.15630 [hep-ph]
5. “Minimal Explanation of Flavor Anomalies: B-Meson Decays, Muon Magnetic Moment, and the Cabibbo Angle”. D. Marzocca, **S. Trifinopoulos**, Phys. Rev. Lett. 127 (2021) 6, 061803 • arXiv: 2104.05730 [hep-ph]
4. “Exploring the flavour structure of the high-scale MSSM”. G. Isidori, **S. Trifinopoulos**, Eur. Phys. J. C 80 (2020) 291 • arXiv: 1912.09940 [hep-ph]
3. “B-physics anomalies: The bridge between R-parity violating Supersymmetry and flavoured Dark Matter”. **S. Trifinopoulos**, Phys. Rev. D 100, 115022 (2019) • arXiv: 1904.12940 [hep-ph]
2. “Revisiting R-parity violating interactions as an explanation of the B-physics anomalies”. **S. Trifinopoulos**, Eur. Phys. J. C 78 (2018) no. 10, 803 • arXiv: 1807.01638 [hep-ph]
1. “Semileptonic B-physics anomalies: A general EFT analysis within $U(2)_n$ flavor symmetry”. M. Bordone, G. Isidori, **S. Trifinopoulos**, Phys. Rev. D 96 (2017) 1, 015038 • arXiv: 1702.07238 [hep-ph]

Large-Collaboration Publications & White Papers

10. “GUEST: Gravitational Universe Exploration with Satellite Tracking. A passive satellite laser-ranging mission for the dark gravitational Universe”. D. Blas et al. (incl. **S. Trifinopoulos**) • arXiv: 2607.18390 [astro-ph.CO, gr-qc]
9. “The Muon Collider”. C. Accettura et al. (incl. **S. Trifinopoulos**) • arXiv: 2504.21417 [physics.acc-ph]
8. “Quantum Information meets High-Energy Physics: Input to the update of the European Strategy for Particle Physics”. Y. Afik et al. (incl. **S. Trifinopoulos**), Eur. Phys. J. Plus 140, 855 (2025) • arXiv: 2504.00086 [hep-ph]

7. “Interim report for the International Muon Collider Collaboration (IMCC)”. C. Accettura et al. (incl. **S. Trifinopoulos**) • arXiv: 2407.12450 [physics.acc-ph]
6. “Towards a Muon Collider”. C. Accettura et al. (incl. **S. Trifinopoulos**), Eur. Phys. J. C 83 (2023) 9, 864 • arXiv: 2303.08533 [physics.acc-ph]
5. “Simulated Detector Performance at the Muon Collider”. Muon Collider Collaboration (incl. **S. Trifinopoulos**), Contribution to Snowmass 2021 • arXiv: 2203.07964 [hep-ex]
4. “A Muon Collider Facility for Physics Discovery”. Muon Collider Collaboration (incl. **S. Trifinopoulos**), Contribution to Snowmass 2021 • arXiv: 2203.08033 [physics.acc-ph]
3. “Promising Technologies and R&D Directions for the Future Muon Collider Detectors”. Muon Collider Collaboration (incl. **S. Trifinopoulos**), Contribution to Snowmass 2021 • arXiv: 2203.07224 [physics.ins-det]
2. “Muon Collider Physics Summary”. C. Aime et al. (incl. **S. Trifinopoulos**), Contribution to Snowmass 2021 • arXiv: 2203.07256 [hep-ph]
1. “The physics case of a 3 TeV muon collider stage”. Muon Collider Collaboration (incl. **S. Trifinopoulos**), Contribution to Snowmass 2021 • arXiv: 2203.07261 [hep-ph]

Workshop Publications and Proceedings

5. “The entangled collider”. A. Barr, **S. Trifinopoulos**, CERN Courier (2026)
4. “r-process Nucleosynthesis: Identifying the significant nuclear properties”. S. G. Tomasicchio, **S. Trifinopoulos**, 2023 REYES Proceedings (<https://digitalcommons.odu.edu/reyes-2023/>)
3. “NuDyCLR: Nuclear Dynamic Co-Learned Representations”. V. S. Pérez-Díaz, **S. Trifinopoulos**, 2023 REYES Proceedings (<https://digitalcommons.odu.edu/reyes-2023/>)
2. “NuCLR: Nuclear Co-Learned Representations”. O. Kitouni, N. Nolte, **S. Trifinopoulos**, S. Kantamneni, M. Williams, accepted after peer review at ICML 2023 “1st Workshop on Synergy of Scientific and Machine Learning Modeling” (SynS & ML) • arXiv: 2306.06099 [nucl-th, cs.lg]
1. “Explaining the Flavour Anomalies with Heavy Scalars”. **S. Trifinopoulos**, PoS CORFU2021 (2022) 052 • arXiv: 2203.09624 [hep-ph]